

JEE Main - 4 | [JEE 2022] (GEN 1)

Date : 2/11/2020

Maximum Marks : 300

Timing : 4 PM - 9 PM

Duration : 3.0 Hrs

General Instructions

1. The test is of **3 hours** duration and the maximum marks is **300**.
2. The question paper consists of **3 Parts** (Part I: **Physics**, Part II: **Chemistry**, Part III: **Mathematics**). Each Part has **two** sections (Section 1 & Section 2).
3. **Section 1** contains **20 Multiple Choice Questions**. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE CHOICE** is correct.
4. **Section 2** contains **5 Numerical Value Type Questions**. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.* (Example: 6, 81, 1.50, 3.25, 0.08)

Marking Scheme

1. **Section – 1:** +4 for correct answer, –1 (negative marking) for incorrect answer, 0 for all other cases.
2. **Section – 2:** +4 for correct answer, 0 for all other cases. There is no negative marking.

Name of the Candidate (In CAPITALS) :

Roll Number :

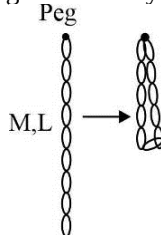
OMR Bar Code Number :

Candidate's Signature : **Invigilator's Signature**

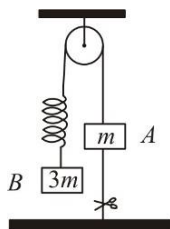
PART I : PHYSICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

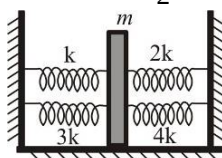
1. What will be the work done by external agent to slowly hang the lower end of the chain to the peg?



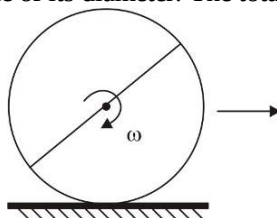
- (A) $-\frac{MgL}{2}$ (B) $\frac{MgL}{2}$ (C) $-\frac{MgL}{4}$ (D) $\frac{MgL}{4}$
2. The system shown in figure is initially in equilibrium. What is the acceleration of A just after the string connecting A to ground is cut?



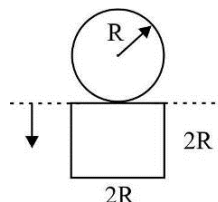
- (A) g downwards (B) $\frac{4g}{3}$ upwards (C) zero (D) $2g$ upwards
3. A circular ring of mass 8kg and radius 2m is placed such that its centre lies at the origin. Two particles of masses 3kg and 4kg are placed at the intersecting points of the circle with '+'ve x -axis and '+'ve y -axis respectively. Then the angle made by the position vector of centre of mass of entire system with positive y -axis is:
- (A) $\tan^{-1}\left(\frac{2}{3}\right)$ (B) $\tan^{-1}\left(\frac{3}{4}\right)$ (C) $\tan^{-1}\left(\frac{4}{5}\right)$ (D) $\tan^{-1}\left(\frac{3}{7}\right)$
4. A block of mass m is attached to four unstretched massless springs of spring constant k , $2k$, $3k$ and $4k$, respectively as shown in the figure. The block is displaced towards right through distance x and is released. Speed of block when displacement of block is $\frac{x}{2}$ from mean position is:



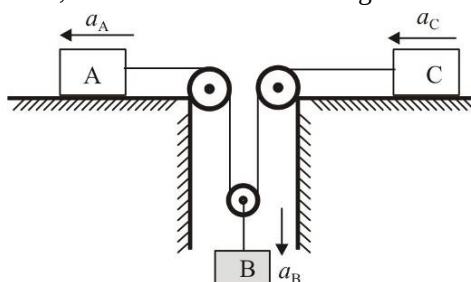
- (A) $x\sqrt{\frac{15m}{2k}}$ (B) $x\sqrt{\frac{15k}{2m}}$ (C) $x\sqrt{\frac{10m}{2k}}$ (D) $x\sqrt{\frac{10k}{2m}}$
5. A disk of mass m and radius R , is rolling without slipping with angular velocity ω as shown in figure. A rod of mass $2m$ is fixed along one of its diameter. The total kinetic energy of the system is:



- (A) $\frac{25}{12}\omega^2 mR^2$ (B) $\frac{19}{12}\omega^2 mR^2$ (C) $1.4\omega^2 mR^2$ (D) $\frac{7}{12}\omega^2 mR^2$
6. There is a uniform plane lamina as shown in the figure. A dotted axis is lying in the plane of the lamina. As the axis is moved downwards, moment of inertia of the lamina about the axis will:
- (A) increase continuously
(B) decrease continuously
(C) first increase then decrease
(D) first decrease then increase



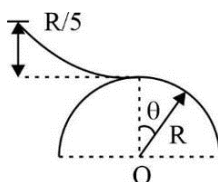
7. Figure shows an arrangement in which three identical blocks are joined together with an inextensible string. All the surfaces are smooth and pulleys are massless. If a_A , a_B and a_C are the respective accelerations of the blocks A, B and C, then which of the following relations is correct?



- (A) $a_A + 2a_B = a_C$ (B) $a_A + 2a_B + a_C = 0$
(C) $a_B = \frac{a_A - a_C}{2}$ (D) $a_A = a_C$
8. A block of mass 4 kg is placed on a rough surface and is being acted upon by a time dependent force $F = 4t^2$ (N) (where t is in second). The co-efficient of friction between the block and horizontal surface is $\mu = 0.4$. Then velocity of block at $t = 5$ s is:

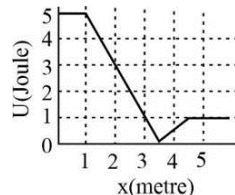


- (A) 27 m/s (B) 50 m/s (C) 22.5 m/s (D) 20 m/s
9. A force starts acting on a particle of mass 1.5 kg so that its position is given as a function of time as $x = (10t^2 + 20)$ metre. What is the work done by this force in first 2 seconds?
- (A) 600J (B) 900 J (C) 750J (D) 1200 J
10. A skier plans to ski a smooth fixed hemisphere of radius R. He starts from rest from a curved smooth surface of height $\left(\frac{R}{5}\right)$. The angle θ at which he leaves the hemisphere is:

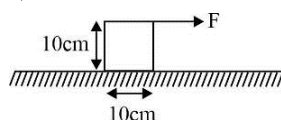


(A) $\cos^{-1}\left(\frac{4}{5}\right)$ (B) $\cos^{-1}\frac{5}{\sqrt{3}}$ (C) $\cos^{-1}\left(\frac{5}{6}\right)$ (D) $\cos^{-1}\left(\frac{3}{5}\right)$

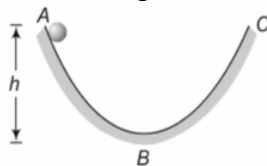
11. A body with mass 10kg moves in one direction in the presence of a force which is described by the potential energy graph. If the body is released from rest at $x = 1\text{m}$, then its speed when it crosses $x = 5\text{m}$ is: (Neglect dissipative forces)?



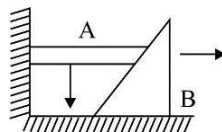
- (A) $2\sqrt{5}\text{ms}^{-1}$ (B) $2\sqrt{2}\text{ms}^{-1}$ (C) $5\sqrt{2}\text{ms}^{-1}$ (D) $\frac{2}{\sqrt{5}}\text{ms}^{-1}$
12. A cubical block of mass $m = 1\text{kg}$ and side $a = 10\text{cm}$ is placed on a sufficiently rough horizontal surface to prevent sliding before toppling. A time varying horizontal force, $F = \left(\frac{5t}{4}\right)\text{N}$, where t is in second, is applied along the perpendicular bisector of an edge on the block's upper surface. At what time the block will topple? ($g = 10\text{ms}^{-2}$)



- (A) 8 sec (B) 6 sec (C) 4 sec (D) None of these
13. A solid ball rolls down a parabolic path ABC from a height h as shown in figure. Portion AB of the Path is sufficiently rough while BC is smooth. How high will the ball climb in BC?



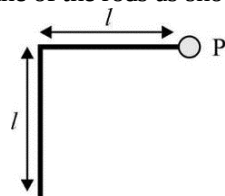
- (A) $\frac{5}{7}h$ (B) $\frac{3}{5}h$ (C) $\frac{1}{2}h$ (D) $\frac{2}{5}h$
14. What will be the force in Newton which mass A exerts on mass B if B is moving towards right with 1ms^{-2} and A is moving downwards with 2ms^{-2} . Mass of B is 1kg . (All surfaces are smooth)



- (A) $\sqrt{10}$ (B) $\frac{\sqrt{10}}{2}$ (C) $\frac{\sqrt{5}}{2}$ (D) None of these
15. A projectile is moving at 20ms^{-1} at its highest point where it breaks into three equal parts A, B and C due to an internal explosion. Just after explosion, A moves vertically up at 80ms^{-1} , B falls from rest, then C will move at:

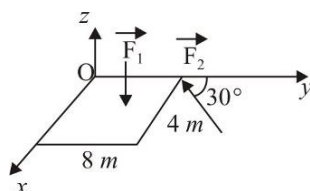
- (A) 80ms^{-1} (B) 100ms^{-1} (C) 20ms^{-1} (D) 60ms^{-1}

16. Find the moment of inertia of the two uniform joint rods having mass m each, about an axis passing through P and perpendicular to the plane of the rods as shown in the figure:

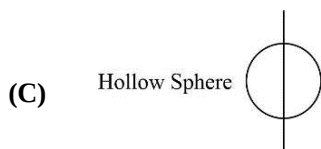
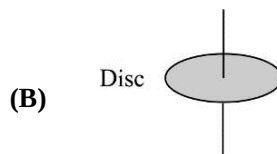
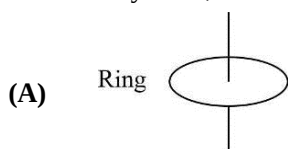


- (A) $\frac{7}{12}ml^2$ (B) $\frac{5}{3}ml^2$ (C) $\frac{7}{3}ml^2$ (D) $\frac{5}{12}ml^2$
17. A uniform bar of mass m and length L is horizontally suspended from the ceiling by two vertical light cables. Cable A is connected at a distance of $\frac{L}{6}$ from the left end of the bar. Cable B is attached at the far right end of the bar. What is the tension in cable A?

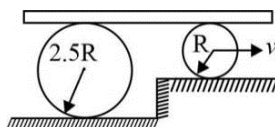
- (A) $\frac{1}{4}Mg$ (B) $\frac{1}{3}Mg$ (C) $\frac{2}{3}Mg$ (D) $\frac{3}{5}Mg$
18. A slab is subjected to two forces $\vec{F}_1$ and $\vec{F}_2$ of same magnitude F as shown in the figure, force $\vec{F}_2$ is in XY -plane while force $\vec{F}_1$ acts along $-z$ axis at the point $(2\hat{i} + 4\hat{j})$. The moment of these forces about point O will be:



- (A) $(4\hat{i} - 2\hat{j} - 4\hat{k})F$ (B) $(4\hat{i} + 2\hat{j} + 4\hat{k})F$
- (C) $(-4\hat{i} + 2\hat{j} + 4\hat{k})F$ (D) $(-4\hat{i} - 2\hat{j} + 4\hat{k})F$
19. For same mass M and same radius R , which one of the following uniform bodies will have minimum radius of Gyration, about its axes as shown:



20. In the given arrangement, the velocity of the small cylinder is v . There is no slipping anywhere. The velocity of the centre of the large cylinder is:



- (A) v (B) $2v$ (C) $\frac{5}{2}v$ (D) None of these

SECTION-2

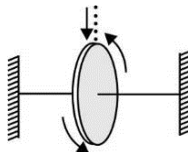
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(Example: 6, 81, 1.50, 3.25, 0.08)

21. A box is dropped on a conveyor belt moving at 3 m/sec. If the coefficient of friction between the belt and the box is 0.6, the displacement of the box relative to the conveyor belt before slipping between the two is stopped is W m. Find W .
22. The speed of a uniform hollow sphere in metre/sec after rolling down an inclined plane of vertical height 12 m from rest without sliding is $\left(g = 10 \text{ m/s}^2\right)$:
23. A small sphere tied to the string of length 0.7 m is describing a vertical circle so that the maximum and minimum tensions in the strings are in the ratio 4:1. What will be the velocity of the sphere (in metre/sec) at the lowest position.
24. A string is wound around a solid cylinder of mass 4 kg and radius 0.5 m. If the string is now pulled with a horizontal force of 60 N, and the cylinder is rolling without slipping on a horizontal surface (see figure), then the angular acceleration of the cylinder will be (Neglect the mass and thickness of the string):
(in rad/sec^2)



25. A disc of mass M rotates freely about a fixed horizontal axis through its centre. A thin cotton pad is fixed to its rim, which can absorb water. The mass of water dripping onto the pad is μ per second. If after time $\frac{15M}{k\mu}$, the angular velocity of the disc gets reduced to one fourth of its initial value, find the value of K .



SPACE FOR ROUGH WORK

PART II : CHEMISTRY**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

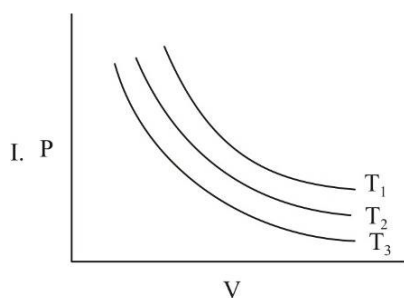
1. When 1 mole of gas is filled in a balloon, the volume of the balloon is 150 mL at 7°C. At this stage, the balloon has been filled to $\frac{1}{3}$ rd of its maximum capacity. Which of the following statements is/are correct?

- I. The balloon will burst at 300°C
 II. The minimum temperature at which the balloon bursts is 500°C

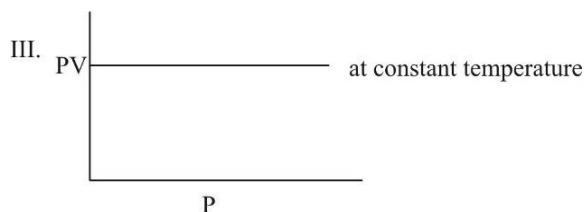
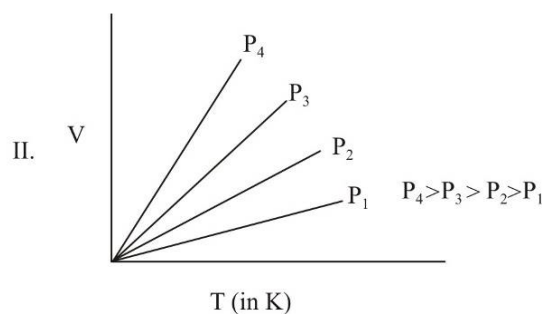
- (A) Only I (B) Only II (C) I and II (D) None of the above
2. A container contains 30 g gas A (molar mass = 60 g – mol⁻¹) and 90 g gas B (molar mass = 45 g – mol⁻¹). The total pressure is 100 mm of Hg. What would the partial pressure (in mm Hg) of gas A be?
- (A) 80 mm of Hg (B) 50 mm of Hg (C) 20 mm of Hg (D) None of the above
3. 1 mol of a gas occupies a volume of 5L at 600K and 3atm. Which of the following statements regarding it is/are correct? (use $R = 0.08 \text{ atm L mol}^{-1} \text{ K}^{-1}$)

- I. The compressibility factor Z for the gas is 5/16.
 II. The volume occupied by the gas at same temperature and pressure will be less than that occupied by an ideal gas.
 III. The gas shows positive deviation

- (A) Only I (B) Only I and II (C) I and III only (D) I, II, III
4. Which of the following graphs is/are correct?



$$T_1 < T_2 < T_3$$



- (A) Only I and II (B) Only III (C) I, II, III (D) Only I and III

5. The density of gas A and gas B is in the ratio 1:2. If the molecular mass of A is 4 times that of B, then ratio of partial pressure of A to that of B will be:
- (A) $\frac{1}{4}$ (B) $\frac{8}{1}$ (C) $\frac{1}{8}$ (D) $\frac{2}{1}$
6. Which of the following statements is/are correct?
- I. The values of the constant "a" increases in the same order as the critical temperature.
- II. $P_c = \frac{8a}{27Rb}$, where P_c is critical pressure
- III. Boyle's temperature > Critical temperature
- (A) Only I (B) Only I and III (C) Only II and III (D) I, II, III
7. Which of the following statements is/are correct about the effect of an increase in temperature on the distribution of molecular speed in a gas?
- I. The area under the distribution curve does not change on increasing the temperature.
- II. The fraction of molecules having most probable speed increases.
- III. The most probable speed increases.
- IV. The fraction of molecules in high speed range increases.
- (A) Only I, II, IV (B) Only II, III (C) Only I, III, IV (D) I, II, III, IV
8. Which of the following statements is/are correct?
- I. BCl_3 is planar but NH_3 is non planar.
- II. SnCl_2 has more covalent character than SnCl_4 .
- III. The octet rule is obeyed by the central atom in SCl_2 .
- (A) Only I (B) Only I and III (C) Only III (D) I, II, III
9. In which of the following option bond angles are correctly matched?
- Bond angle: 120° 106.6° 110.9°
- (A) $\text{N}(\text{SiH}_3)_3$ $\text{N}(\text{CH}_3)_3$ NH_3
- (B) $\text{N}(\text{CH}_3)_3$ NH_3 $\text{N}(\text{SiH}_3)_3$
- (C) $\text{N}(\text{SiH}_3)_3$ NH_3 $\text{N}(\text{CH}_3)_3$
- (D) $\text{N}(\text{CH}_3)_3$ $\text{N}(\text{SiH}_3)_3$ NH_3
10. Which of the following is correct order of ionic character?
- (A) $\text{BaCl}_2 > \text{ZnCl}_2 > \text{CaCl}_2 > \text{RbCl}$ (B) $\text{RbCl} > \text{CaCl}_2 > \text{BaCl}_2 > \text{ZnCl}_2$
- (C) $\text{RbCl} > \text{BaCl}_2 > \text{CaCl}_2 > \text{ZnCl}_2$ (D) $\text{ZnCl}_2 > \text{CaCl}_2 > \text{BaCl}_2 > \text{RbCl}$
11. In which of the following, B – F bond length is minimum?
- (A) BF_2OH (B) BF_2NH_2 (C) $\text{BF}_3 \cdot \text{NMe}_3$ (D) BF_3

12. Which of the following statements is/are correct?

I. OF_4 and OF_6 do not exist

II. The dipole moment of  is equal to zero

III. The bond angle of ClO_2 is greater than that of OCl_2

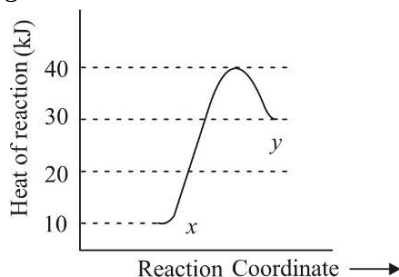
(A) Only I and II

(B) Only I and III

(C) Only III

(D) I, II, III

13. From the graph given below, the value of ΔH for conversion from X to Y is:



(A) -40

(B) -20

(C) 20

(D) 30

14. $2\text{NO}(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) + 2X \text{ kJ}$

$\text{N}_2(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{NO}_2(\text{g}) + 3Y \text{ kJ}$

The enthalpy of formation of NO is:

(A) $2X - 3Y$

(B) $\frac{1}{2}(X - Y)$

(C) $X - \frac{3Y}{2}$

(D) $\frac{3}{2}Y - X$

15. If you are provided with the following data:

Resonance energy of $\text{C}_6\text{H}_6(\text{l}) = -141 \text{ kJ/mol}$

Enthalpy of hydrogenation of $\text{C}_6\text{H}_6(\text{l})$ into $\text{C}_6\text{H}_{12}(\text{l}) = -225 \text{ kJ/mol}$

Calculate the enthalpy of hydrogenation of 1,4-cyclohexadiene(l) to $\text{C}_6\text{H}_{12}(\text{l})$.

(A) +244 kJ/mol

(B) -244 kJ/mol

(C) -238 kJ/mol

(D) +128 kJ/mol

16. The reaction $\text{C}_2\text{H}_6(\text{g}) + \text{Cl}_2(\text{g}) \rightarrow \text{C}_2\text{H}_5\text{Cl}(\text{g}) + \text{HCl}(\text{g})$; $\Delta H = -50 \text{ kcal}$

The bond energies are as follows:

$\text{B.E}_{\text{C-Cl}} = 84 \text{ kcal}$

$\text{B.E}_{\text{H-Cl}} = 103 \text{ kcal}$

$\text{B.E}_{\text{C-H}} = a \text{ kcal}$

$\text{B.E}_{\text{Cl-Cl}} = b \text{ kcal}$

$\text{B.E}_{\text{C-C}} = 70 \text{ kcal}$

$a : b = 9 : 5$

From the given data, the bond energy of Cl-Cl bond (kcal) is:

(A) 49

(B) 55

(C) 70

(D) 45

17. When NH_4OH reacts with HCl , the enthalpy of neutralisation reaction is -50 kJ/mol . Assuming that the enthalpy of neutralisation when a strong acid reacts with a strong base is -57 kJ/mol . What would be the enthalpy of ionisation of NH_4OH ?

(A) -107 kJ/mol

(B) 107 kJ/mol

(C) -7 kJ/mol

(D) 7 kJ/mol

18. Which of the following molecule will not exist in normal conditions?
 (A) XeF_6^{2+} (B) PCl_3 (C) PH_5 (D) TeCl_3^+
19. 320 g of O_2 is inhaled by a person per day. If all the O_2 is used for converting sucrose, $(\text{C}_{12}\text{H}_{22}\text{O}_{11})$ into CO_2 and H_2O , how much heat is evolved per day?
 ΔH combustion of sucrose = -5600kJ/mol . [Atomic mass: O = 16 g/mol]
 (A) -56000kJ (B) -4666.67kJ (C) $-1915 \times 10^3\text{kJ}$ (D) -6720kJ
20. Which of the following statement is/are correct?
 I. The heat of solution is defined as the enthalpy change when one mole of solute is dissolved in excess of solvent so that further dilution does not bring any heat change at constant temperature and pressure.
 II. Heat exchange between system and surrounding at constant volume is equal to change in internal energy.
 III. Density, molar conductance and molarity are intensive properties.
 (A) Only I and III (B) Only III (C) Only II and III (D) I, II, III

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.

(Example: 6, 81, 1.50, 3.25, 0.08)

21. How many of the following species is/are paramagnetic?
 NO , O_2 , B_2 , O_2^{2-} , NO^+ , C_2 , N_2
22. The bond order of P – O bonds in PO_4^{3-} is _____
23. When 0.36 gm of glucose ($\text{C}_6\text{H}_{12}\text{O}_6(\text{s})$) was burnt in a bomb calorimeter (Heat capacity 600 J/K), the temperature rise by 10 K. The standard molar enthalpy of combustion of glucose is $-x \times 10^6 \text{ J/mol}$. Find the value of x.
 [Given: Molecular mass of glucose = 180 g/mol]
24. 1 mol of Gas A (molar mass = $64\text{g} \cdot \text{mol}^{-1}$) escapes from a container through a hole in 8 seconds. A mixture of 0.8 mol gas A and 0.2 mol gas B, escapes from an identical container through same hole (at same T and P) in 9 seconds. What is the molar mass of gas B?
25. An ideal gas molecule has an average speed of 500 m/s at 1207°C . What will be its average speed at 97°C ?

PART III : MATHEMATICS**100 MARKS****SECTION-1**

This section contains 20 Multiple Choice Questions. Each question has 4 choices (A), (B), (C) and (D), out of which ONLY ONE CHOICE is correct.

- If a vertex of a triangle is $(1,1)$ and the middle points of two sides through it are $(-2,3)$ and $(5,2)$ then the centroid of the triangle is:
 (A) $\left(\frac{5}{3}, 3\right)$ (B) $\left(-\frac{5}{3}, 3\right)$ (C) $\left(\frac{5}{3}, -3\right)$ (D) $\left(-\frac{5}{3}, -3\right)$
- The ratio in which the line $3x + 4y - 7 = 0$ divides the line joining the points $(-2,1)$ and $(1,2)$ is:
 (A) 2:3 (B) 4:5 (C) 7:3 (D) 9:4
- Consider all the permutations of the word "JANLOKPAL". Number of words in which O never lies between J and N is equal to:
 (A) $6|7$ (B) $8|7$ (C) $9|7$ (D) $12|7$
- Coefficient of x^5 in the expansion $\left(x + \frac{1}{x}\right)^{10} \left(x - \frac{1}{x}\right)^9$ is:
 (A) 9C_3 (B) 9C_4 (C) $-{}^9C_3$ (D) $-{}^9C_4$
- Given the family of lines, $a(3x + 4y + 6) + b(x + y + 2) = 0$ where a and b are parameters. The line of the family situated at the greatest distance from the point $P(2,3)$ has the equation:
 (A) $4x + 3y + 8 = 0$ (B) $5x + 3y + 10 = 0$
 (C) $15x + 8y + 30 = 0$ (D) None of these
- The value of expansion $\sum_{r=0}^{20} (-1)^r \cdot \frac{{}^{20}C_r}{3^r}$ is:
 (A) $\left(\frac{3}{2}\right)^{20}$ (B) $2^{40} \cdot 3^{20}$ (C) $\left(\frac{2}{3}\right)^{20}$ (D) $\frac{3^{20}}{2^{40}}$
- The point $(-4,5)$ is the vertex of a square and one of its diagonal is $7x - y + 8 = 0$. Then the equation of other diagonal is:
 (A) $7x - y = 23$ (B) $x + 7y = 13$ (C) $7x - y = 31$ (D) $x + 7y = 31$
- The image of point (α, β) in the line $y = -x$ is:
 (A) (β, α) (B) $(-\alpha, -\beta)$ (C) $(-\beta, -\alpha)$ (D) None of these
- The equation of a straight line passes through $(-3,2)$ and cutting an intercept equal in magnitude but opposite in sign from the axes is:

- (A) $x - y + 5 = 0$ (B) $x + y - 5 = 0$ (C) $x - y - 5 = 0$ (D) None of these
10. If the term independent of x in the expansion of $\left(\frac{x^2}{2} - \frac{1}{x}\right)^9$ is k , then $2k$ is equal to:
 (A) 7 (B) 21 (C) 15 (D) 5
11. Coefficient of x^6 in expansion of $\left(x + \sqrt{x^2 - 1}\right)^6 + \left(x - \sqrt{x^2 - 1}\right)^6$, is:
 (A) 60 (B) 62 (C) 64 (D) 66
12. A library has books of 5 different subjects. Assume that sufficient numbers of books in all subjects are available for selection and books of same subject are identical. The number of ways a student can select 7 books is: (Student can select any number of books of a subject)
 (A) 8C_2 (B) ${}^{11}C_7$ (C) 6C_4 (D) 9C_4
13. If $2p$ is length of perpendicular from the origin to line $\frac{x}{a} + \frac{y}{b} = 1$, then:
 (A) $a^2, 8p^2, b^2$ are in A.P. (B) $a^2, 8p^2, b^2$ are in G.P.
 (C) $a^2, 8p^2, b^2$ are in H.P. (D) None of these
14. Straight line $y = x - 2$ is rotated about a point where it cuts x -axis and becomes parallel to the line $5x + 3y - 2 = 0$, then area bounded by the new line with co-ordinates axes, is:
 (A) $\frac{2}{15}$ sq. units (B) 4 sq. units (C) $\frac{10}{3}$ sq. units (D) $\frac{4}{3}$ sq. units
15. The coefficient of x^5 in the expression of $(1 + x^2)^3 (1 + x^4)^7 (1 - x)^3$ is:
 (A) -30 (B) -33 (C) -21 (D) -12
16. If P, Q and R are three points with co-ordinates $(1, 4), (4, 5)$ and (m, m) respectively, then the value of ' m ' for which $PR + RQ$ is minimum is :
 (A) 4 (B) 3 (C) $\frac{17}{8}$ (D) $\frac{7}{2}$
17. Sum of series ${}^{10}C_0 - {}^{10}C_1 + \frac{{}^{10}C_2 2^2}{3} - \frac{{}^{10}C_3 2^3}{4} + \dots + \frac{{}^{10}C_{10} 2^{10}}{11}$ is:
 (A) $-\frac{2}{11}$ (B) $-\frac{1}{11}$ (C) $\frac{1}{11}$ (D) $\frac{2}{11}$
18. The remainder when 27^{40} is divided by 12 is:
 (A) 3 (B) 7 (C) 9 (D) 11
19. If in triangle ABC , $A \equiv (1, 10)$, circumcentre $\equiv \left(-\frac{1}{3}, \frac{2}{3}\right)$ and orthocenter $\equiv \left(\frac{11}{3}, \frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is:
 (A) $\left(1, -\frac{11}{3}\right)$ (B) $(1, 5)$ (C) $(1, -3)$ (D) $(1, 6)$
20. Two vertices of a triangle are $(4, -3)$ and $(-2, 5)$. If the ortho centre of the triangle is at $(1, 2)$ then the third vertex is:
 (A) $(-33, -26)$ (B) $(33, 26)$ (C) $(26, 33)$ (D) None of these

SECTION-2

This section has Five (05) Questions. The answer to each question is a NUMERICAL VALUE. For each question, enter the correct numerical value of the answer. If the answer is a decimal numerical value, then round-off the value to TWO decimal places. If the answer is an Integer value, then do not add zero in the decimal places. *In the OMR, do not bubble the $\oplus$ sign for positive values. However, for negative values, $\ominus$ sign should be bubbled.*

(Example: 6, 81, 1.50, 3.25, 0.08)

21. An equilateral triangle has each of its sides of length 6 cm. If (x_1, y_1) , (x_2, y_2) and (x_3, y_3) are the vertices then the value of the determinant, $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2$ is equal to:
22. If the middle term of $\left(\frac{1}{x} + x \sin x\right)^{10}$ is equal to $\frac{63}{8}$, then find the number of values of x in $[0, 2\pi]$.
23. The natural number P , for which the coefficient of x^{60} in the binomial expansion of $\left(x^P + \frac{1}{x^2}\right)^{25}$ is 2300, is:
24. The number of integers greater than 5000 that can be formed using 3, 4, 5, 6, 0 without repetition is:
25. An eight digit number is formed from 1, 2, 3, 4 such that product of all digits is always 3072, the total number of ways is $\left(k \cdot {}^8C_5\right)$, where the value of k is: